🚲 BIKE SHARING DATA ANALYSIS (Power BI PROJECT REPORT)

1. Project Overview

This project focuses on analyzing a **Bike Sharing System** using Power BI to evaluate city-wise performance, operational efficiency, utilization patterns, and growth trends.

The goal is to transform raw operational data into meaningful insights that support **urban mobility planning and infrastructure optimization**.

2. Objective

The main objective of this analysis is to:

- Understand bike utilization patterns across cities
- Evaluate operational and revenue efficiency
- Identify high-performing and underperforming cities
- Analyze seasonal and usage trends
- Provide data-driven recommendations for system improvement

3. Problem Statement

The project aims to solve the following business problems:

- Evaluate city-wise performance and revenue efficiency
- Identify top and low-performing cities
- Analyze usage trends and seasonal patterns
- Detect capacity mismatch and inefficiencies
- Support infrastructure planning and expansion decisions

4. Dataset Description

The dataset contains operational details of bike-sharing stations across multiple cities.

Attribute	Description
Station ID	Unique identifier for each station
Name	Station name

Attribute	Description
Address	Location details of station
Position	Latitude and Longitude coordinates
Banking	Indicates payment facility availability
Bonus	Indicates reward/bonus availability
Status	Operational status (Open/Closed)
Contract Name	City or operating region
Bike Stands	Total station capacity
Available Bike Stands	Empty docking spaces
Available Bikes	Bikes currently available
Last Updates	Latest data refresh timestamp

5. Tools & Technologies

- **Power BI** → Dashboard creation & visualization
- **DAX (Data Analysis Expressions)** → Calculations & KPIs
- **Power Query** → Data cleaning and transformation
- **Excel/CSV Dataset** → Data source

6. Data Pre-Processing

✓ Data Cleaning

- Removed duplicate records
- Handled missing values
- Standardized formats
- Fixed inconsistent station names

✓ Data Transformation

- Converted raw data into Fact and Dimension tables

- Created calculated fields for analysis

✓ **Filtering & Structuring**

- Focused on relevant operational metrics
- Organized data for city-wise and time-based analysis

7. Data Modeling

- Built **star schema model** using Fact and Dimension tables
- Established relationships between station, city, and usage data
- Enabled efficient filtering and slicing in Power BI

8. Analysis Performed

◆ **Descriptive Analysis**

- High demand observed in cities like Lyon, Bruxelles-Capitale, Toulouse, and Valencia
- Bike utilization rate is **34%**, indicating underutilized capacity
- Uneven distribution of bike stands across cities

◆ **Diagnostic Analysis**

- Overall cost efficiency is **0.66**, indicating strong financial performance
- Cities like Vilnius, Rouen, and Marseille show high efficiency
- Higher capacity does not always lead to higher utilization

◆ **Predictive Analysis**

- System recorded **3,503 bikes in use last year**
- Overall YoY growth rate is **9.72%**
- Cities like Nancy, Maribor, and Toulouse show strong future growth potential

◆ **Prescriptive Analysis**

- High-performing cities should focus on expansion and service enhancement
- Mid-performing cities require redistribution and optimization
- Low-performing cities need demand stimulation and performance monitoring

9. Key Insights

- 🚲 Average utilization is only **34%**, indicating unused capacity
- 🚲 Strong demand is concentrated in few major cities
- 🚲 Cost efficiency remains high despite uneven usage
- 🚲 Significant growth potential exists (9.72% YoY increase)
- 🚲 Infrastructure imbalance exists across cities

10. Business Recommendations

- Expand bike infrastructure in high-performing cities
- Redistribute bikes to balance demand
- Optimize underperforming stations
- Improve marketing in low-usage regions
- Monitor capacity utilization regularly

11. Dashboard Features

- KPI Cards (Total Bikes, Utilization Rate, Growth Rate)
- City-wise Performance Analysis
- Utilization vs Capacity Comparison
- Growth Trend Analysis
- Interactive Filters (City, Status, Time)

12. Conclusion

The analysis reveals that the bike-sharing system is financially stable and growing, but suffers from uneven utilization across cities. While some cities show high demand and efficiency, others remain underutilized.

Strategic expansion, redistribution, and targeted optimization can significantly improve system performance and ensure sustainable growth.

13. Skills Demonstrated

- Power BI Dashboard Design

- Data Cleaning & Transformation
- Data Modeling (Fact & Dimension tables)
- DAX Calculations
- Business Intelligence Storytelling
- Operational & Financial Analysis